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A dust generator has been developed to produce dry dust from powdery materials for instrument calibration and laboratory studies. The powder is fed at a constant rate via a chain conveyor into a fluidized bed where the particles are deagglomerated and aerosolized. The powder feed rate is variable from 1.2 to 36 mm³/min and the air flow rate from 9 to 30 Lpm. Aerosols of coal, silica, potash, rock, and Arizona road dust have been successfully generated for calibrating optical particle counters and aerosol mass monitors.

A dust generator for laboratory use

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introduction

Laboratories engaging in particle research often must generate aerosols from substances initially in a powder form. For example, dust generation is often required in order to test dust measuring and sampling instruments. Dust generation is also needed in many animal exposure experiments in which the toxicity of inhaled dust particles is assessed. For these studies, it is usually desirable that the powder be thoroughly dispersed, and that all particles appear in the gas stream as single particles, rather than as agglomerates of two or more particles. It is also desired that the size distribution and concentration of the dust aerosol be constant with respect to time.

A variety of methods have been used to generate dust aerosols.⁽¹⁾ If good particle dispersion and constant concentration are unimportant, then such simple methods as blowing a dust from a tube with a blast of air, or stirring of the dust in a blender, can be used. However, to obtain stable operations, more sophisticated dust generators, such as the Wright dust generator⁽²⁾ or the sonic jet dust feeder⁽³⁾ can be used. Although these generators give a constant feed rate over long periods of time, the instantaneous feed rate may vary. Better feed rate control can be obtained by metering the dust before it is dispersed, such as in the turntable dust feeder.⁽⁴⁾ However, in all of these generators, the generated dust particles usually are not thoroughly deagglomerated.

Another technique of aerosolizing dust particles, which does thoroughly deagglomerate the dust particles, is by the use of a fluidized bed. In this technique, the powder to be dispersed is introduced into a fluidizing bed of beads which are larger than the powder, and the fluidizing action of these beads disperses the powder as individual particles. Several dust generators using this principle have been developed and are described in the literature,^(5,6) and it is this principle of dust dispersion that is used in the dust generator described in this paper.

the laboratory dust generator

The dust generator described in this paper has been developed as a general purpose generator for laboratory use. The initial application has been for calibrating dust measuring instruments such as optical particle counters^(7,8) and mass monitors.⁽⁹⁾ The volume flow rate needed is relatively low, but the particles must be thoroughly deagglomerated, and the concentration and size distributions of the aerosol must remain constant. To meet these requirements, the dust is aerosolized by a fluidized bed to which powder is fed continuously by a chain conveyor mechanism.

The dust generator developed is shown in Figures 1 and 2. The powder to be aerosolized is placed in a holding chamber, whence it is transported by a chain conveyor into the

fluidized bed. The fluidized bed consists of a 1.5 cm thick layer of 100 μm brass beads placed in a 5.1 cm diameter chamber. The beads are supported by a 250 mesh nylon screen placed above an air plenum. The fluidizing air flows from the plenum through the fluidized bed, in which the powder is dispersed and carried by the fluidizing air through the elutriation chamber. The air velocity in the elutriator is high enough to transport the dust particles, but not high enough to transport the brass beads, which fall back into the bed. Also located in the elutriation chamber is a Kr-85 radioactive source to reduce the particle electrical charge to the Boltzmann equilibrium level.

The dust generated in this fashion is referred to as the "total" aerosol in this paper. It is also possible to use only the "respirable" portion of the aerosol by placing an optional 1/2-inch HASL cyclone at the top of the elutriation tube, and operating the fluidized bed at a flow rate of 9 Lpm.⁽¹⁰⁾ As the aerosol passes through this cyclone, the large nonrespirable particles will be removed in accordance with the ACGIH respirable curve.

A critical part of this dust generator is the chain conveyor system, which feeds the powder into the fluidized bed at a constant rate. The conveyor is a continuous no. 3 ball chain, (2.3 mm diameter balls on 3.1 mm centers) which passes between the dust holding chamber and the fluidized bed. In the dust holding chamber, the chain, which is driven by a sprocket, picks up powder in the space between the balls as it passes through the powder. The chain enters the fluidized bed through the lower hole connecting the two chambers. This hole is lined with a resilient material to scrape the excess powder from the chain before entering the fluidized bed, and to form a seal preventing the fluidized brass beads from entering the powder holding chamber. The chain then passes over a Teflon skid on the opposite side of the fluidizing chamber and returns to the holding chamber through a second hole, located above the fluidized bed.

To insure that the chain does not create a channel in the powder, and thus fail to pick up uniform amounts of material between the balls, a rake in the bottom of the holding chamber is connected to the chain drive mechanism to

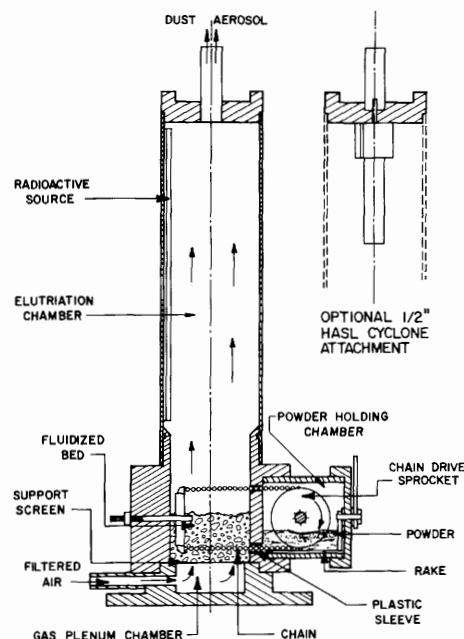
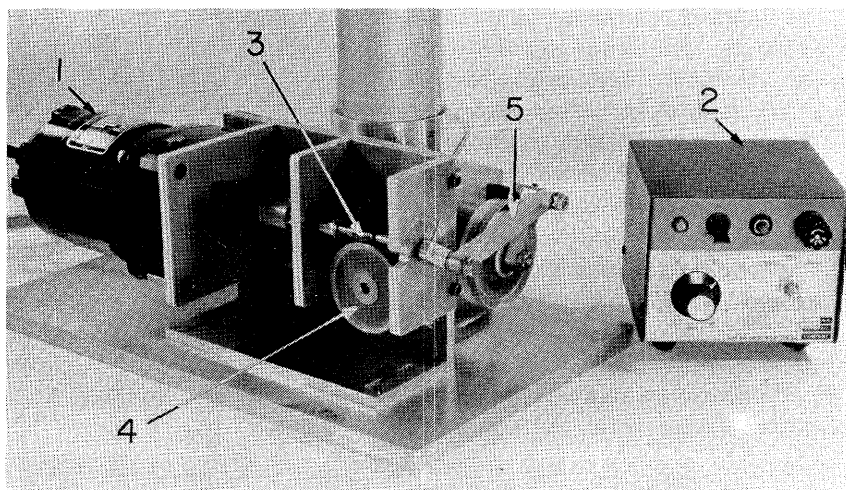


Figure 1 — Dust generator.

continuously stir the dust around the chain. This stirring action makes it impossible for a channel to form in the powder, thus insuring a constant powder feed rate.

The drive mechanism, shown in Figure 2, is powered by a variable speed gearmotor (Bodine Model NSH-11D4). The motor rotates a shaft which drives the chain sprocket shaft through a worm-worm gear arrangement, and also drives the stirring rake through a four-bar linkage.

After operating the dust generator for approximately 1000 hours, with several types of dust, the generator was disassembled and the parts checked for wear. The only parts which were found to show any appreciable wear were the Teflon skid and the lining in the hole between the dust holding chamber and the fluidized bed. The ball chain was found to have no measurable wear. From previous inspections of the dust generator, it was known that most of the wear of the lining and skid was caused by the generation of silica dust, which was very abrasive. The lining, which was a soft rubber, was replaced with a polypropylene lining. In subsequent tests, this lining was found to be more resistant to wear. However, there has not been enough operating time on a new lining to determine how often it must be replaced.



b. Side view

a. Top view (with elutriation chamber removed)

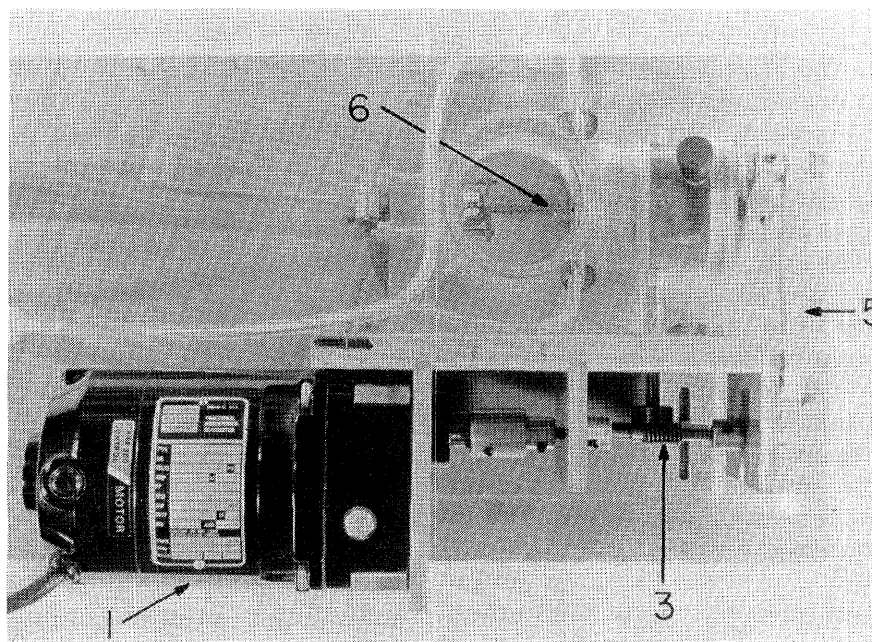


Figure 2 — Conveyor chain and stirring rake drive mechanism.

- | | |
|----------------------------|---|
| 1. Gearmotor | 5. Four-bar linkage stirring rake drive mechanism |
| 2. Gearmotor speed control | 6. Conveyor chain |
| 3. Worm | |
| 4. Worm gear | |

The effect of the worn lining on the operation of the dust generator was to increase the dust feed rate by about 10%. The only effect from the worn skid was on the chain tension, which can be easily tightened by adjusting the bracket holding the skid.

generator performance

A schematic of the test setup for evaluating the dust generator's performance is shown in Figure 3. Aerosol from the generator flows through a 180° arched tube. Dilution air is added, and the diluted aerosol then flows downward through a

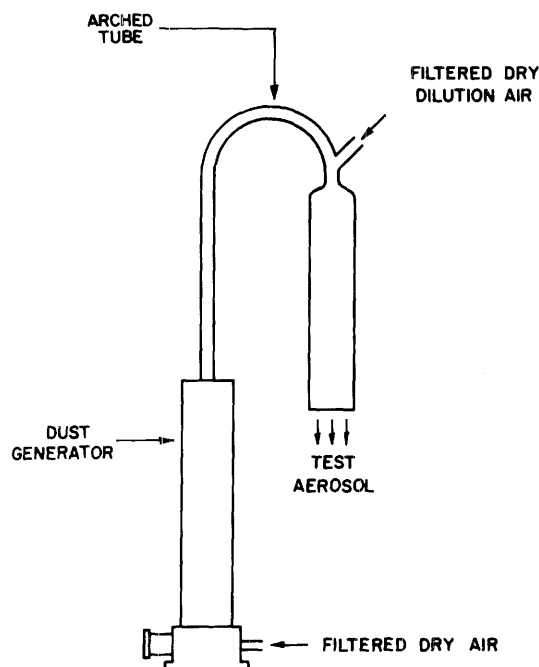


Figure 3 — Test setup for evaluating dust generator.

5.1 cm diameter tube. At the exit of this tube, the aerosol is sampled for generator evaluation.

Several instruments were used to evaluate the dust generator. The mass concentration was measured by collecting the particles on a filter, and measuring the increase in filter weight. The mass and number size distributions were determined by the use of a cascade impactor and an optical particle counter, respectively.

The bed was found to fluidize best at air flow rates in the range of 9 to 30 Lpm. The corresponding range of maximum particle sizes which can pass through the elutriator for this range of flow rates is from 45 μm to 85 μm aerodynamic diameter. However, even at 9 Lpm

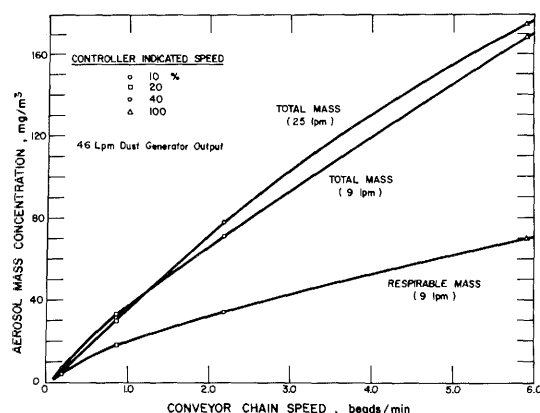


Figure 4 — Dust generator output (Arizona road dust - AC Fine).

the maximum elutriated particle aerodynamic diameter of 45 μm is of sufficient size for most studies.

The mass concentration of the dust aerosol generated is controlled primarily by the speed of the conveyor chain. The speed of the motor can be varied between 3% and 100% of the maximum speed by a variable speed controller. This corresponds to a chain speed range from 0.2 to 6 chain balls/min.

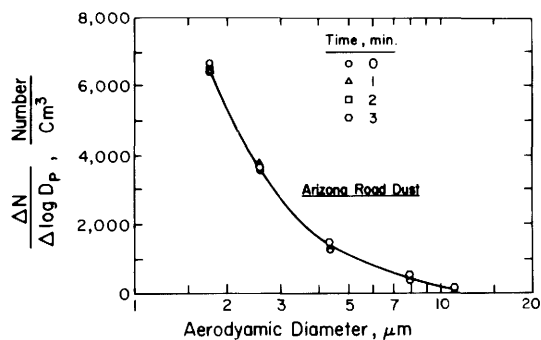
The maximum volume of material that can be fed to the fluidized bed by each chain ball is 6.7 mm³. However, as shown in Table I for the aerosol materials that have been studied, the volumetric feed rate is more nearly 6 mm³/ball and is fairly independent of the material. The volumetric feed rate in Table I was calculated from the mass feed per ball (measured gravimetrically) and the bulk density of each material.

The mass concentration of the output aerosol from the generator is shown in Figure 4 for

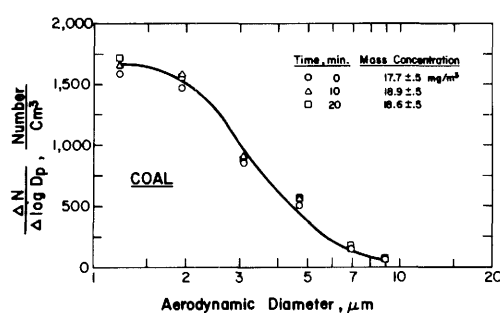
TABLE I
Conveyor Chain Feed Characteristics

Material	Particle Density (gm/cc)	Bulk Density (gm/cc)	Mass Feed (mg/ball)	Calculated Volumetric Feed* (mm ³ /ball)
Coal	1.45	0.50	3.3	6.6
Potash	2.14	0.70	4.2	6.0
Arizona Road Dust	2.61	0.83	4.6	5.5
Silica	2.62	1.02	5.3	5.2
Rock (copper ore)	4.61	1.12	6.3	5.6

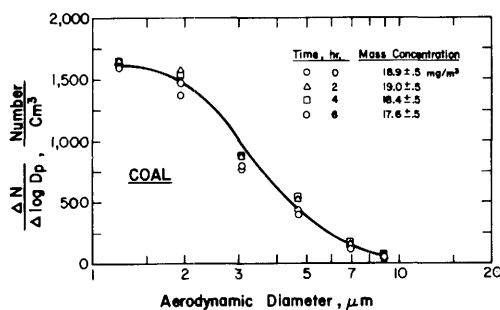
*Actual volumetric feed = 6.66 mm³/ball.



a, one-minute intervals



b, ten-minutes intervals



c, two-hour intervals

Figure 5 — Dust generator stability studies.

Arizona road dust (AC Fine). For a chain speed of 0.2 to 6 balls/min, the "total" mass concentration ranges from 5 to 175 mg/m³ at a flow rate of 46 Lpm. Since the flow rate through the fluidized bed was 9 or 25 Lpm, as shown in Figure 4, the dilution air had to be varied to make up a total flow rate of 46 Lpm at the output. The "total" and "respirable" curves refer to whether or not the aerosol was passed through

the cyclone. The concentration for the "respirable" aerosol is about one-half of that of the "total" aerosol. Also note in Figure 4 that the output concentration is nearly the same for both 9 and 25 Lpm of flow in the fluidized bed, indicating that the same amount of particles were aerosolized in both cases.

The output aerosol mass as calculated from the "total" mass concentration shown in Figure 4 is about 35% of the mass which was conveyed into the fluidized bed by the chain-feed mechanism as determined from the data in Table I. Although some of the particles were too large to be aerosolized, most of the losses are known to be in the arched tube at the top of the elutriation chamber. Thus, if the dust was taken directly from the elutriation tube, the mass concentration would be considerably higher than that shown in Figure 4.

To study the stability of the dust generator, an optical particle counter which had been calibrated by inertial impactors⁽⁷⁾ was used to monitor the generated dust. The resulting size distributions, shown in Figures 5a, 5b, and 5c, demonstrate the degree of stability for one-minute, ten-minute, and two-hour intervals, respectively.

In Figure 5a, the size distribution of the Arizona road dust is shown for four consecutive one-minute runs. The resulting size distributions are seen to be nearly identical. The stability of the generator over longer periods of time is demonstrated in Figures 5b and 5c for coal dust. Each of the curves in Figures 5b and 5c represents run times of 5 and 15 minutes, respectively. The mass concentration in each figure, was determined gravimetrically by weighing filter

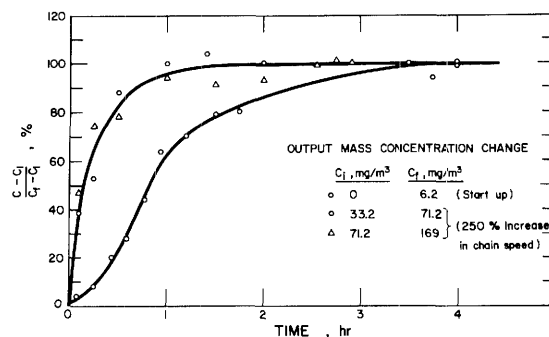


Figure 6 — Time requirements for reaching steady state.

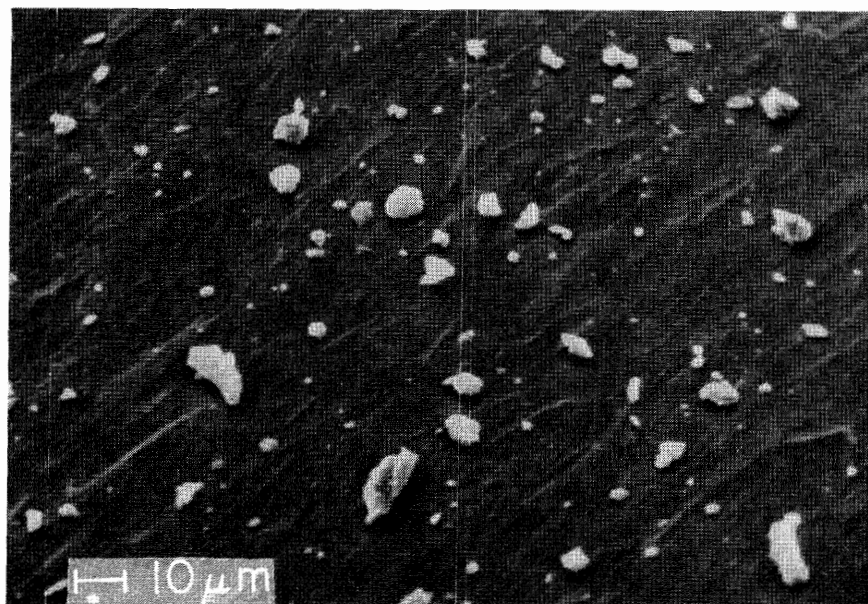


Figure 7 — Coal particles generated by dust generator.

samples. The uncertainty value on the mass concentration was based on estimates of the uncertainty of the mass flow rate, and time measurements of the collected filter samples.

The data in Figures 5a, 5b, and 5c was taken after the dust generator was operating for sufficient time at the given flow rate and chain speed to reach steady operation. The minimum time it takes to reach stable operating conditions is shown in Figure 6. Here, the mass concentration history as a function of time is shown for periods of time after step changes in the chain speed have been made. This figure shows essentially two types of curves. One curve represents a startup case, where the fluidized bed initially contains only clean brass beads with zero mass concentration output. For this case, it takes on the order of three hours for the fluidized bed to reach stable operation. The other case is when the fluidized bed chain conveyor is increased from one to another speed. Two sets of data points are shown for this case, both of which have had an increase in chain speed of about 250%. For these cases, stable operation at the new concentration has been reached in about one hour's time.

The instantaneous fluctuations in the concentrations of the dust being generated by the generator were observed by passing

generated coal dust through a photometer. Although the photometer was not calibrated to give mass concentrations of the generated dust, it did show cyclic variations in the concentration with a frequency which corresponded to the frequency of the beads entering the fluidized bed. These fluctuations were on the order of $\pm 5\%$ of the average concentration. The average concentration, however, was found to be steady.

Besides the particle size distribution and concentration stability of the dust generator, another important property of this dust generator is its ability to thoroughly deagglomerate the aerosolized particles. This characteristic was analyzed by collecting dust particles from the fluidized bed at low concentrations and examining them under a microscope. For all materials generated to date, which include coal (shown in Figure 7), Arizona road dust, silica, rock (copper ore), and potash, the particles have been thoroughly deagglomerated. For these dusts, Table II lists the mass mean aerodynamic diameter (MMD) and geometric standard deviation (σ_g) for both the "total" and "respirable" dust mass size distributions. For all materials, except the coal dust, the size distributions were found to be log-normal. The σ_g value for the coal dust was taken as D_{p50}/D_{p18} . The size distributions for coal,

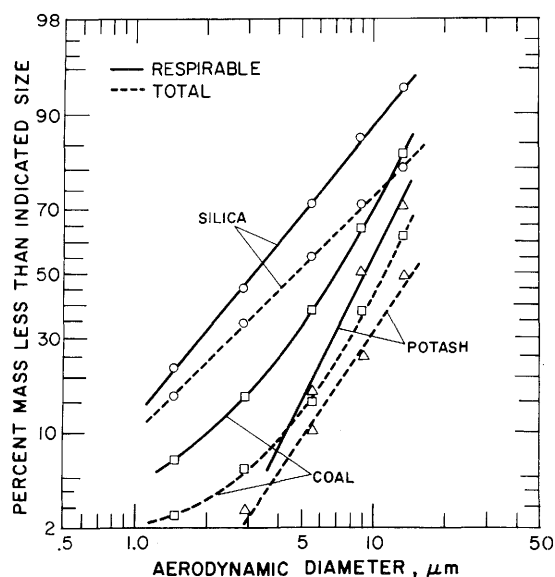


Figure 8 — Typical aerosol size distributions of particles dispersed by the dust generator.

potash, and silica dust are shown in Figure 8. The mass size distributions shown for these dusts were determined by using a cascade impactor made up of 5 stages of the calibration impactor described in reference 7. Grease coated glass coverslips were used on the impaction plates for easy gravimetric analysis of the dust deposits.

It should be noted that whenever there was a change in the material to be aerosolized, the old powder and fluidized bed beads were removed from the generator, the generator cleaned, and new fluidized bed beads and powder added to the dust generator. However, the removed powder and fluidized bed beads were saved and reused when the dust generator was again used to generate dust of that material.

conclusions

A dust generator has been designed and developed using a chain conveyor to feed powder into a fluidized bed dust dispersion system. Aerosols of coal, Arizona road dust, silica, rock (copper ore), and potash particles have been generated, and the dust particles were found to be thoroughly deagglomerated. Tests have shown that the particle concentration and size distribution of the generated aerosol is constant with respect to time. The output concentration from this generator for a typical powder (Arizona road dust) can be varied from

TABLE II
Dust Generator Mass Size Distribution

Material	Total Aerosol		Respirable Aerosol	
	MMD (μm)	σ_g	MMD (μm)	σ_g
Coal	11.0	2.7	7.2	2.2
Potash	14.5	2.2	9.2	1.8
Arizona Road Dust	6.5	2.6	4.4	2.3
Silica	4.7	3.4	3.1	2.7
Rock (copper ore)	8.1	3.3	4.2	2.0

near 0 (by the use of dilution air) to 170 mg/m^3 at a flow rate of 46 Lpm, which corresponds to a mass output of 8 mg/min.

acknowledgement

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